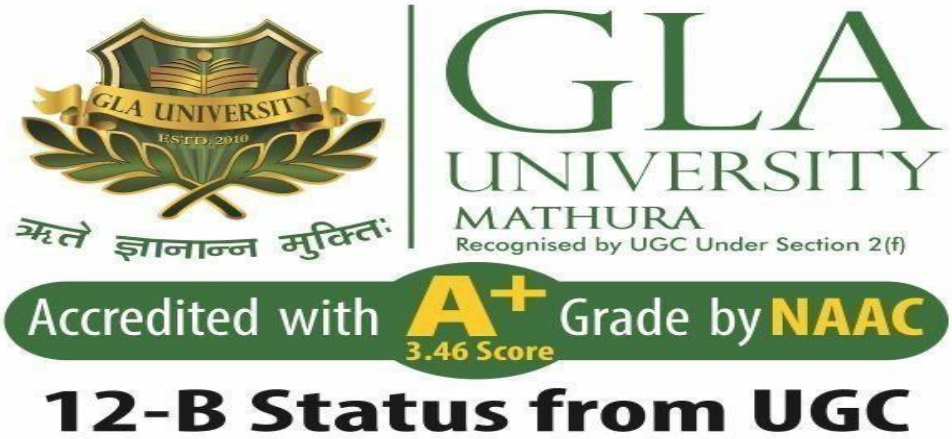
B.Tech CSE (III YEAR – V SEM) (2026-27)

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING & APPLICATIONS



GLA UNIVERSITY

17km Stone, NH-2, Mathura-Delhi Road, P.O. Chaumuhan, Mathura – 281406
(Uttar Pradesh) India

Project Title: AI-Powered Valet Transparency and Vehicle Security Platform

Team Member 1: Mahesh UR: 2415500275

Team Member 2: Jagesh Goyal UR: 2415500209

Mentor Name: Dr. Gourav Bathla

Signature :

Project Synopsis:

AI-Powered Valet Transparency and Vehicle Security Platform

Cover

- **Project title:** AI-Powered Valet Transparency and Vehicle Security Platform
  Team name & ID: DOOM (T-54)
- **Institute / Course:** GLA University, Mini-Project
- **Version:** v1.0
- **Date:** 03 Sep 2026

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1. Overview

- **Problem statement:** Valet parking provides convenience to customers at hotels, restaurants, shopping centers, and other venues. However, once customers hand over their vehicle keys, they have limited visibility into how their vehicle is handled.
- **Goal:** Develop an AI-powered valet platform that uses computer vision for vehicle damage detection, geospatial and ML techniques for suspicious movement detection, and predictive analytics for vehicle retrieval-time estimation.
- **Non-goals:** Autonomous vehicle control, automatic legal/insurance decisions, full OEM connected-car integration, commercial POS integration, and production-level IoT hardware are outside the scope of Version 1.
- **Value proposition:** A transparent and intelligent valet experience providing digital vehicle tickets, AI-based condition reports, security alerts, vehicle-location visibility, and predicted retrieval times to improve customer trust and valet operational efficiency.

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2. Scope and Control

2.1 In-scope

- Vehicle registration and digital valet ticket generation
- AI-based scratch and dent detection
- Before/after vehicle condition comparison
- GPS/telemetry monitoring and geofencing
- ML-based suspicious movement detection
- Vehicle retrieval-time prediction
- Customer and admin dashboards
- Security alerts and vehicle reports

2.2 Out-of-scope

- Autonomous vehicle control
- Automatic legal/insurance decisions
- OEM connected-car integration in V1
- Real POS integration in V1
- Physical IoT hardware in the initial prototype
- Dedicated mobile application in V1

2.3 Assumptions

- Vehicle images are available for inspection.
- GPS/telemetry data can be obtained or simulated.
- Parking boundaries can be defined.
- Historical valet data is available or can be generated.

2.4 Constraints

- Limited real-world valet datasets
- Variations in image quality and lighting

- GPS accuracy limitations
- Limited project timeline and resources

2.5 Dependencies

- Vehicle damage datasets
- GPS/telemetry data
- Historical valet records
- YOLO, OpenCV and Scikit-learn
- Database and mapping services

2.6 Acceptance criteria and sign-off

- **GIVEN** a vehicle is registered **WHEN** check-in is completed **THEN** a digital valet ticket is generated.

- **GIVEN** vehicle images are uploaded **WHEN** AI processes them **THEN** scratches and dents are detected.

- **GIVEN** telemetry is received **WHEN** a vehicle leaves the geofence **THEN** a security alert is generated.

- **GIVEN** a retrieval request is made **WHEN** relevant data is available **THEN** the system predicts retrieval time.

Sign-off table

Stakeholder	Role	Decision area	Signature/Approval	Date
Dr. Gourav Bathla	Mentor	Scope, final acceptance	Approved	03 Sep 2026
Mahesh	Team Leader	Project readiness	Approved	03 Sep 2026

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Mahesh	Mahesh	Mentor	Jagesh Goyal
AI/ML Design	Mahesh	Mahesh	Mentor	Jagesh Goyal
System Design	Jagesh Goyal	Mahesh	Mentor	Team
Implementation	Mahesh & Jagesh Goyal	Mahesh	Mentor	Team
Testing	Jagesh Goyal	Mahesh	Mentor	Team
Documentation	Mahesh & Jagesh Goyal	Mahesh	Mentor	Team
Release	Mahesh & Jagesh Goyal	Mahesh	Mentor	Department

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4. Team and Roles

Member	Role	Responsibilities	Key Skills	Availability	Contact
Mahesh	Team Leader & AI/ML Lead	Project planning, AI/ML models, vehicle damage detection, security analysis	Python, AI/ML, Computer Vision, OpenCV	10 hrs/wk	mahesh.gla_cs.aiml24@gla.ac.in
Jagesh Goyal	Full-Stack Developer	Backend, database, dashboard, GPS/geofencing integration, system testing	Python, Flask/FastAPI, SQL, HTML/CSS/JS	10 hrs/wk	jagesh.goyal.gla_cs.aiml24@gla.ac.in

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5. Week-wise Plan and Assignments

Week	Dates	Milestone	Mahesh (Team Leader & AI/ML)	Jagesh Goyal (Development)	Deliverables	Status
1	25 Aug 2026	Requirements & Planning	Finalize scope, requirements and project objectives	Study system requirements and technology stack	SRS, Project Plan	Planned
2	03 Sep 2026	System Architecture & Data	Design AI/ML workflow and data requirements	Design architecture, database and APIs	Architecture, ER Diagram	Planned
3	10 Sep 2026	Vehicle Damage Detection	Prepare dataset and develop damage detection model	Create image upload and database modules	CV Model, Image Module	Planned
4	15 Sep 2026	Security Monitoring	Develop GPS, geofencing and suspicious movement logic	Integrate GPS data, maps and security APIs	Security Module	Planned
5	25 Sep 2026	Retrieval Prediction	Develop retrieval-time prediction model	Implement ticket, queue and retrieval interfaces	Prediction Module	Planned
6	10 Oct 2026	Dashboard & Integration	Integrate AI results and generate alerts/reports	Develop customer and admin dashboards	Working Dashboard	Planned
7	20 Oct 2026	Testing & Evaluation	Evaluate AI/ML accuracy and improve models	Perform system, API and integration testing	Test Report, Improved Models	Planned
8	5 Nov 2026	Final Release & Presentation	Final AI/ML validation, documentation and presentation	Final integration, UI polishing and deployment	Final Project, PPT, User Manual	Planned

6. Users and UX

6.1 Personas

User	Needs
Vehicle Owner / Customer	Track vehicle status, view location, check vehicle condition and receive security alerts
Valet Operator	Register vehicles, update parking status and manage retrieval requests
Valet Manager	Monitor vehicles, security alerts, damage reports and retrieval performance

6.2 Top User Journeys

Customer →

Vehicle Check-in →

Digital Ticket →

View Vehicle Status/Location →

Vehicle Condition Report →

Security Monitoring →

Request Retrieval →

Vehicle Pickup

6.3 User Stories

- As a vehicle owner, I want to view my vehicle's status and location so that I know where my vehicle is.
- As a vehicle owner, I want an AI-based condition report so that I have evidence of my vehicle's condition.
- As a vehicle owner, I want security alerts for suspicious movement so that I can identify unauthorized vehicle use.
- As a valet operator, I want to register vehicles digitally so that parking records are easy to manage.
- As a valet manager, I want to monitor alerts and retrieval times so that I can improve valet security and efficiency.

6.4 Accessibility & Localization

Requirement	**Implementation**
Simple interface	Clear labels and easy navigation
Mobile-friendly	Responsive customer and operator dashboard
Visual alerts	Clear status and security notifications

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7. Market and Competitors

7.1 Competitor Table

Competitor / Product	Target Users	Key Features	Pricing	Strengths	Limitations / Gap
NimblePark	Hotels, malls, restaurants, hospitals, valet operators	Digital valet tickets, vehicle tracking, valet assignment, retrieval management, real-time dashboard, OTP-based retrieval	Contact vendor	Complete valet operation management	Does not emphasize AI-based vehicle damage detection and ML-based suspicious movement analysis
ValetPlus	Indian hotels, restaurants and businesses	Digital ticketing, real-time car tracking, automated payments, customer notifications	Contact vendor / app-based service	Designed for digital valet operations in India	Limited evidence of integrated AI damage comparison and predictive retrieval analytics
PUR Valet	Hotels and hospitality businesses	AI-assisted vehicle check-in, SMS ticketing, vehicle tracking, retrieval requests, damage documentation, payments and PMS integration	Contact vendor	Strong guest experience and digital valet workflow	Focuses primarily on valet operations; proposed system combines this with security analytics and ML-based movement detection
Inspektlabs	Automotive, insurance, rental and fleet businesses	AI vehicle inspection, scratch/dent detection, image analysis, damage reports and photo-quality checks	Contact vendor	Specialized AI vehicle damage detection	Not a complete valet-management and vehicle-security platform
Ajjas GO	Vehicle owners	Real-time GPS tracking, location monitoring, overspeed alerts and geofence alerts	₹3,790 listed price	Dedicated vehicle tracking and security device	Requires tracking hardware and is not designed around the complete valet check-in, damage inspection and retrieval workflow

7.2 Positioning

Our **AI-Powered Valet Transparency and Vehicle Security Platform** focuses on **vehicle transparency, security, damage detection and efficient retrieval**.

Key Differentiators:

- AI-based scratch and dent detection
- Before/after vehicle condition comparison
- GPS and geofencing-based security monitoring
- ML-based suspicious movement detection
- Predictive vehicle retrieval time
- Digital valet ticket and customer dashboard
- Centralized security and damage reports

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8. Objectives

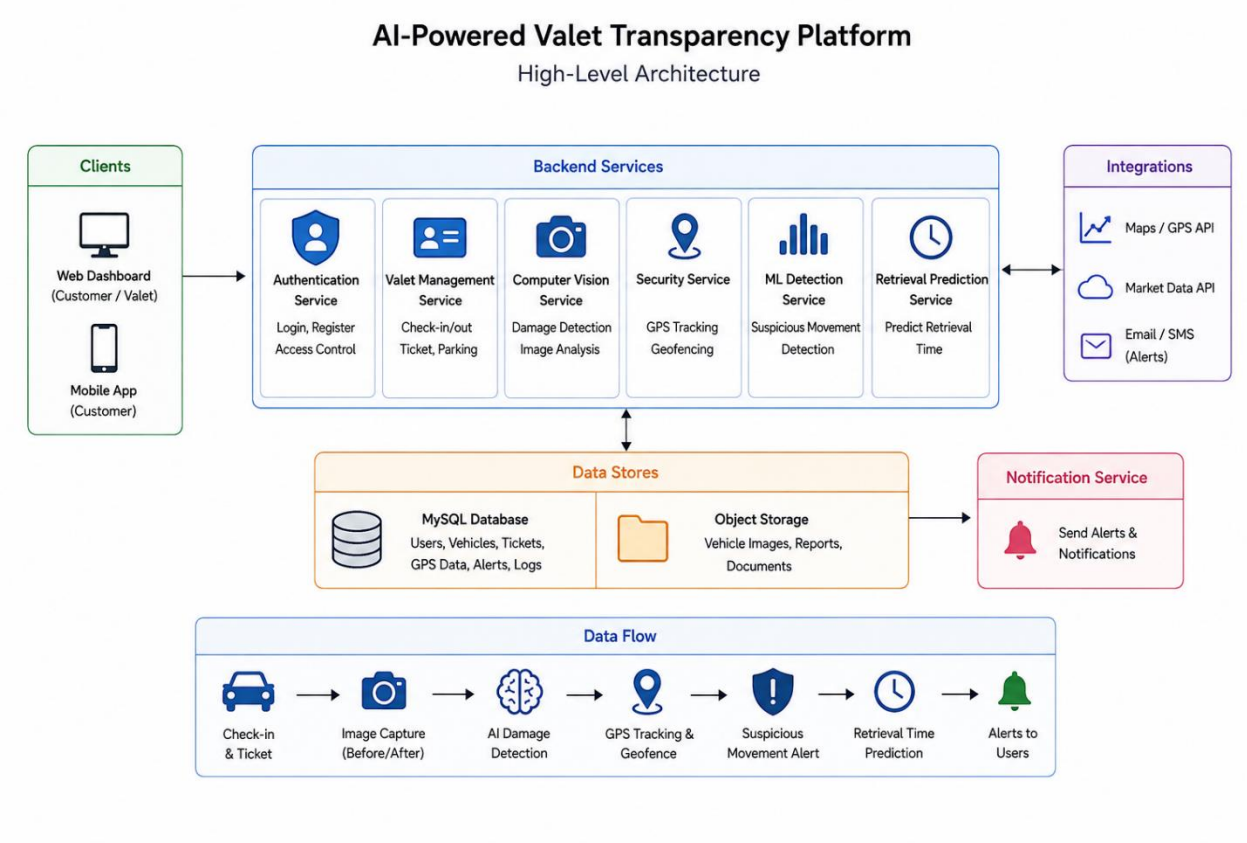
Objective	Specific Objective	Key Problem Solved
O1: Digital Valet Management	To develop a digital vehicle check-in and valet ticket system that records vehicle and customer details and provides a traceable valet transaction.	Lost/manual tickets and poor record management
O2: AI-Based Damage Detection	To develop a computer-vision system that detects visible scratches and dents from vehicle images and compares check-in and check-out conditions.	Manual inspection errors and vehicle-damage disputes
O3: Vehicle Security Monitoring	To monitor vehicle GPS/telemetry data using geofencing and ML-based analysis to identify unauthorized movement, abnormal speed and route deviation.	Lack of visibility and delayed detection of suspicious vehicle movement
O4: Intelligent Retrieval Prediction	To predict the estimated vehicle retrieval time using valet queue information and historical/synthetic retrieval data.	Uncertain waiting time and inefficient vehicle retrieval
O5: Unified Transparency Dashboard	To provide customers and valet managers with a centralized dashboard showing vehicle status, location, condition reports, security alerts and retrieval information.	Lack of transparency and centralized information for customers and valet operators

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9. Key Features

Feature	Description	Priority	Dependencies	Acceptance Criteria
Digital Valet Ticket	Creates a digital ticket containing vehicle and customer details at check-in.	Must	Database, Authentication	GIVEN valid vehicle details WHEN check-in is completed THEN a unique digital ticket is generated.
AI Vehicle Damage Detection	Detects scratches, dents and visible damage from vehicle images.	Must	OpenCV, YOLO, Image Dataset	GIVEN a vehicle image WHEN it is uploaded THEN detected damages are identified and recorded.
Before/After Comparison	Compares check-in and check-out images to identify new or changed vehicle damage.	Must	Damage Detection, Image Storage	GIVEN before and after images WHEN comparison is performed THEN differences are highlighted in the damage report.
GPS & Geofencing	Tracks vehicle location and checks whether it remains within the permitted parking area.	Must	GPS/Telemetry, Maps	GIVEN vehicle location data WHEN the vehicle crosses the defined geofence THEN a security alert is generated.
Suspicious Movement Detection	Identifies abnormal speed, route deviation and unauthorized movement using telemetry data.	Must	GPS Data, ML Model	GIVEN vehicle telemetry WHEN abnormal movement is detected THEN the system generates a suspicious-activity alert.
Retrieval Time Prediction	Predicts estimated vehicle retrieval time using queue and historical valet data.	Should	Historical Data, ML Model	GIVEN a retrieval request WHEN prediction is performed THEN an estimated retrieval time is displayed.
Customer Dashboard	Shows valet ticket, vehicle status, location, condition report and retrieval status.	Must	Authentication, Backend API	GIVEN a customer logs in WHEN the dashboard loads THEN current vehicle information is displayed.

10. Architecture



10.1 High-Level Architecture

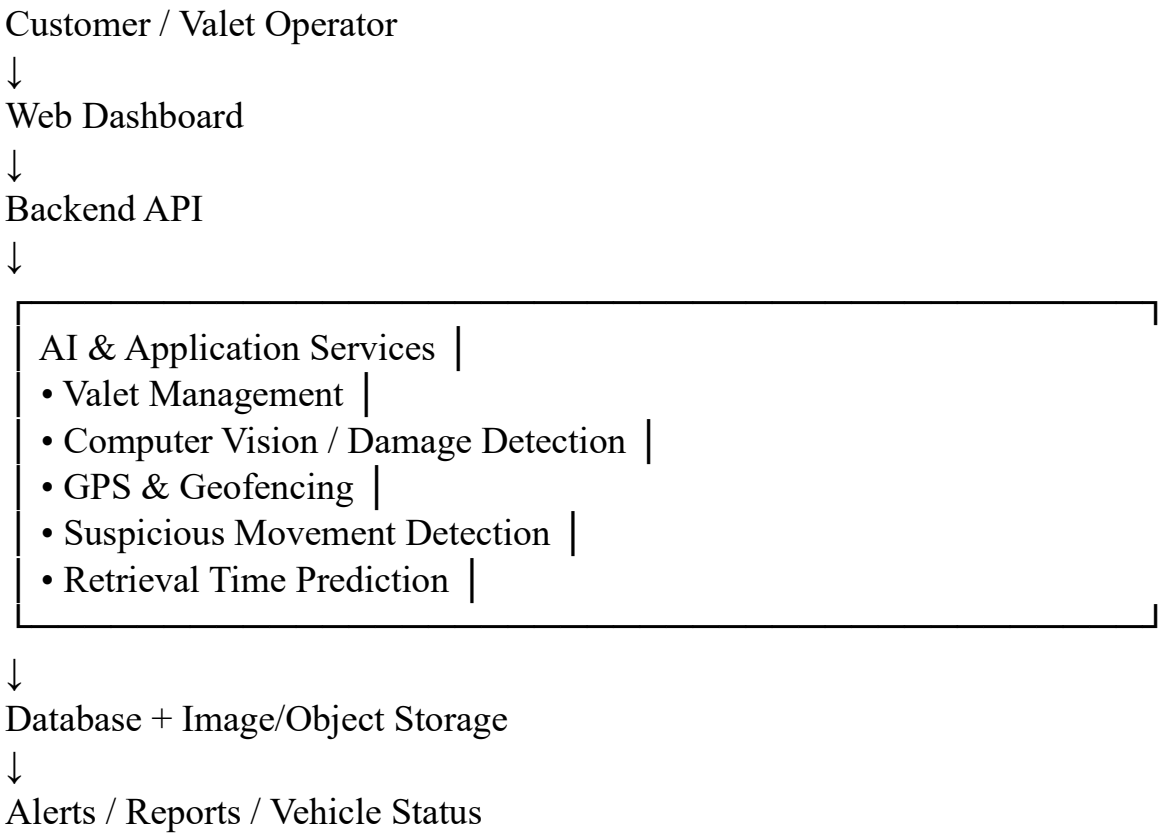
The system follows a **modular architecture** designed to provide vehicle condition monitoring, security tracking, and efficient valet management.

System Flow:

Customer / Valet Operator → Web Dashboard → Backend API → AI/ML Services → Database & Storage → Alerts / Reports

Component	Description
Clients	Web-based responsive dashboard for customers, valet operators and managers.
Authentication Service	Handles user registration, login and role-based access.
Valet Management Service	Manages vehicle check-in, digital tickets, parking status and retrieval requests.
Computer Vision Service	Analyzes vehicle images to detect scratches, dents and other visible damage.
Security & GPS Service	Processes GPS/telemetry data, geofencing, speed and route deviations.
ML Detection Service	Detects suspicious vehicle movement and potential unauthorized use.
Retrieval Prediction Service	Predicts estimated vehicle retrieval time using queue and historical data.
Database	Stores users, vehicles, tickets, inspections, GPS records, alerts and retrieval data.
Object Storage	Stores vehicle images, damage reports and related documents.
Notification Service	Sends security alerts, status updates and retrieval notifications.
Maps/GPS Integration	Provides vehicle location and geofence visualization.

Architecture Diagram



10.2 API spec snapshot

Endpoint	Method	Auth	Purpose	Request Schema	Response Schema	Codes
/api/auth/register	POST	—	Create account	name, email, password	{id}	201, 400
/api/vehicles/checkin	POST	JWT	Register vehicle & create ticket	vehicleNo, customerId, images	{ticketId}	201, 400, 401
/api/vehicle/damage	POST	JWT	Analyze vehicle image	image, ticketId	{damages[]}	200, 400

/api/security/track	POST	JWT	Process GPS/telemetry	vehicleId, latitude, longitude, speed	{status, alert}	200, 400, 401
/api/retrieval/predict	GET	JWT	Predict retrieval time	ticketId	{estimatedTime}	200, 404
/api/alerts	GET	JWT	View security alerts	vehicleId	{alerts[]}	200, 401

10.3 Configuration and Secrets

Security Practice	Implementation
Environment variables	API keys and database credentials stored in .env
Secret protection	.env files excluded using .gitignore
Authentication	JWT-based secure access
Role-based access	Separate customer, valet and manager permissions
API security	Authorized requests only
Data protection	Vehicle images and personal data stored securely
Logging	Security events and important system activities recorded

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11. Data Design

11.1 Core Entities

Entity	Purpose	Key Data
User	Stores customer, valet and manager accounts	User ID, name, email, role
Vehicle	Stores vehicle information	Vehicle ID, registration number, model
Valet Ticket	Tracks the complete valet transaction	Ticket ID, vehicle ID, check-in/out time, status
Vehicle Inspection	Stores vehicle condition information	Inspection ID, images, detected damage
GPS Telemetry	Stores vehicle movement data	Location, speed, timestamp, route
Security Alert	Records suspicious activity	Alert type, severity, location, timestamp
Retrieval Request	Manages vehicle retrieval	Request ID, ticket ID, request time, status
Retrieval Prediction	Stores predicted retrieval time	Prediction, actual time, error
Damage Record	Stores detected scratches/dents	Damage type, location, confidence score

11.2 Schemas and Migrations

Item	Details
ER Diagram	Included in the appendix
Database Migrations	Version-controlled database schema changes
Rollback	Rollback procedures tested before deployment

11.3 Privacy, Retention, Backup, and DR

Item	Details
PII	Customer name, email and vehicle-related information
Data Retention	Inactive customer data retained/deleted according to project policy
Backup	Automated database and file backups
RTO	4 hours
RPO	24 hours
Data Security	Access control and encryption for sensitive data

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12. Technical Workflow Diagrams

Diagram	Purpose
State Transition Diagram	Shows vehicle states such as Check-in → Parked → Retrieval Requested → Ready → Picked Up.
Sequence Diagram	Shows interaction between Customer, Valet Operator, Backend, AI/ML Services and Database.
Use Case Diagram	Shows system functions for Customer, Valet Operator and Manager.
Data Flow Diagram (DFD)	Shows the flow of vehicle, image, GPS and alert data through the system.

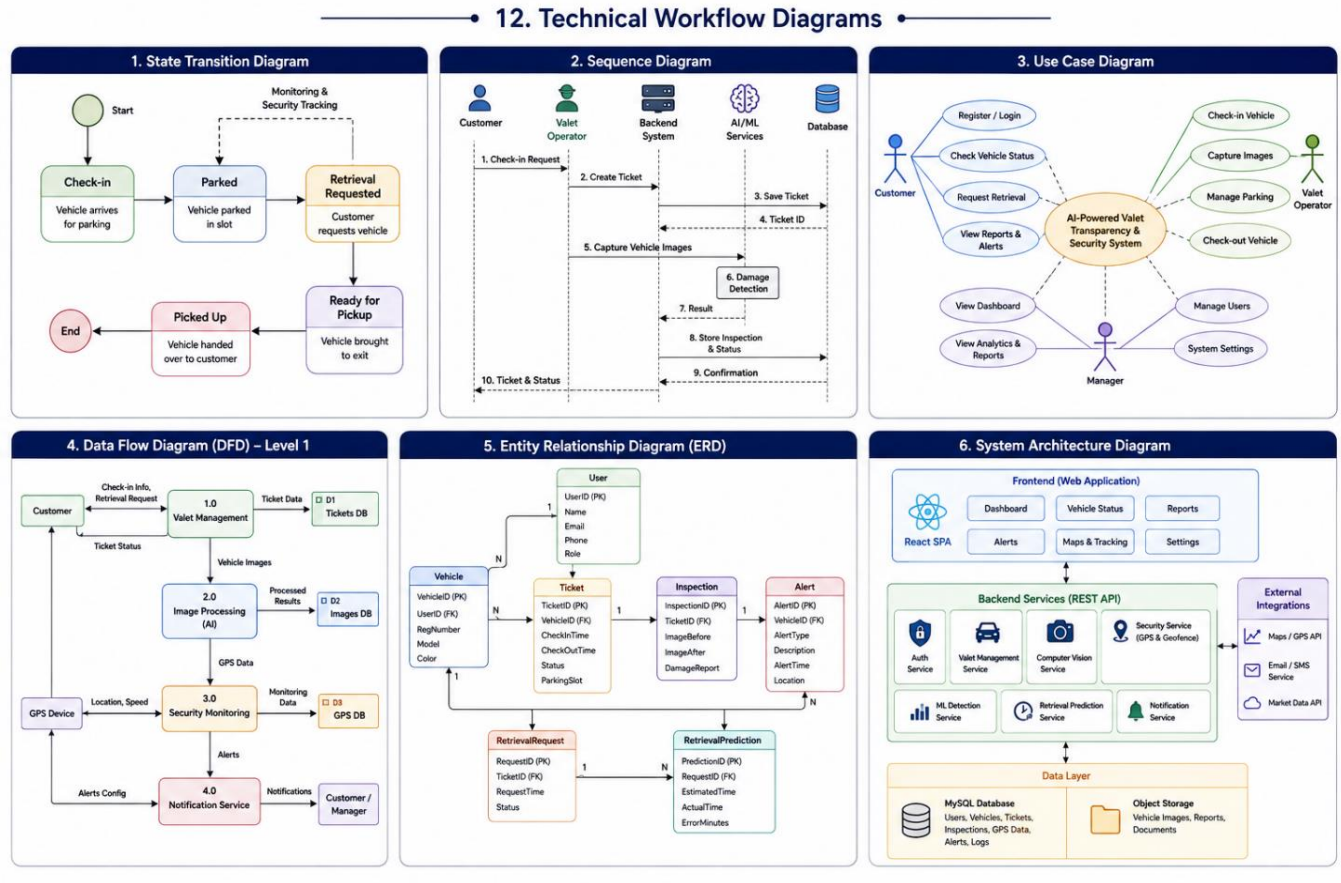
Entity Relationship Diagram (ERD)	Shows relationships between User, Vehicle, Ticket, Inspection, GPS Data, Alerts and Retrieval records.
System Architecture Diagram	Shows Frontend, Backend, AI/ML services, Database, Storage and external integrations.

Overall Workflow:

Vehicle Check-in →Image Capture →AI Damage Detection →

GPS Tracking →Security Monitoring →Retrieval Request →

ETA Prediction →Vehicle Pickup



13. Quality: NFRs and Testing

13.1 Non-functional requirements

Metric	SLI	Target (SLO)	Measurement
Availability	Uptime %	≥ 99.0%	Uptime monitoring
API Latency	p95 response time	≤ 2 seconds	API monitoring
Error Rate	5xx %	≤ 1%	Application logs
AI Detection Accuracy	Precision/Recall	≥ 85%	Model evaluation
Security	Critical vulnerabilities	0	Vulnerability scanner
Data Security	Unauthorized access	0 incidents	Security logs

13.2 Test plan

Area	Type	Tools	Owner	Coverage Target	Exit Criteria
Backend	Unit Testing	PyTest	Jagesh	≥ 70%	No critical defects
AI/ML	Model Testing	Python, Scikit-learn, OpenCV	Mahesh	All core scenarios	Accuracy target achieved
Computer Vision	Functional Testing	OpenCV, YOLO	Mahesh	Major damage cases	Damage detection works correctly
API	Integration Testing	Postman	Jagesh	All critical APIs	All critical tests pass
UI	Functional/E2E	Playwright	Jagesh	≥ 80%	Pass rate ≥ 95%
Security	Security Testing	OWASP checks	Mahesh & Jagesh	Critical flows	No critical vulnerabilities
System	End-to-End Testing	Browser + Postman	Both Members	Complete workflow	Check-in to pickup works successfully

13.3 Environments

Environment	Purpose
Development	Feature development and initial testing
Staging	Integration and final validation
Production/Demo	Final project demonstration and deployment

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14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation	Owner
User Authentication	Token theft / unauthorized login	Spoofing	High	Medium	HTTPS, JWT expiry, secure password hashing	Mahesh
User & Vehicle Data	Unauthorized modification / SQL Injection	Tampering	High	Low	Parameterized queries, input validation, access control	Jagesh Goyal
Vehicle Images	Unauthorized access	Information Disclosure	High	Medium	Access control, secure storage, HTTPS	Jagesh Goyal
GPS / Location Data	Data manipulation	Tampering	High	Medium	Validation, authenticated APIs, audit logs	Mahesh
Security Alerts	False or unauthorized alerts	Spoofing / Tampering	Medium	Low	Authentication, role-based access, logging	Mahesh
System APIs	Excessive requests / service disruption	Denial of Service	High	Medium	Rate limiting, input validation, monitoring	Jagesh Goyal

14.2 AuthN/AuthZ

Security Control	Implementation
Authentication	Email + password
Authorization	JWT-based authorization
Access Control	Role-based access control (Customer, Valet, Manager)
Password Security	Secure password hashing
Communication	HTTPS for data transmission

14.3 Audit and logging

- Log **signups, logins, vehicle check-in/out, image analysis, security alerts and retrieval requests**.
- Maintain logs for **90 days**.
- Restrict access to security and audit logs.
- Record important security events for troubleshooting and investigation.

14.4 Compliance

Area	Requirement
Project Type	Academic project
Data Usage	Use only required customer, vehicle and telemetry data
Privacy	Protect personal and vehicle-related information
Third-Party Sharing	No unauthorized third-party data sharing
Institute Policy	Follow applicable GLA University project and data policies

15. Delivery and Operations

15.1 Release Plan

The project will be developed incrementally, with each major module tested before integration. The final v1.0 version will be prepared for project submission and demonstration.

- v1.0: Final working prototype at project submission
- Features will be integrated and tested incrementally.
- Major changes will be reviewed before the final release.

15.2 CI/CD and rollback

A simple CI/CD workflow will be followed:

Code Push → Lint → Test → Build → Deploy

If a deployment introduces a critical issue, the system can be rolled back to the previous stable version.

15.3 Monitoring and alerting

Metric	Threshold	Alert To	Runbook
p95 API Latency	> 2000 ms	Jagesh Goyal	Check API/DB performance
Error Rate	> 2%	Jagesh Goyal	Inspect logs and rollback
AI Detection Failure	> 10% failed requests	Mahesh	Check model/service
Security Alerts	Unusual spike	Mahesh & Jagesh	Review GPS and security logs

15.4 Runbooks

API Latency: Check API and database performance → identify bottleneck → optimize/scale → redeploy if required.

Error Spike: Inspect application logs → identify faulty component → fix or rollback → record the incident.

AI Model Failure: Check input images/data → verify model service → review prediction errors → retrain or replace model if required.

15.5 Communication plan

- **Team meetings:** Monday, Wednesday and Friday
- **Mentor update:** Weekly
- **Progress demo:** Every two weeks
- **Final demo:** At project submission/presentation

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16. Risks and Mitigations

16.1 Risk heatmap

Risk	Probability	Impact	Score	Mitigation	Owner	Status
Schedule Delay	Medium	High	12	Clear milestones, weekly progress reviews	Mahesh	Open
Limited Dataset	High	High	15	Use public datasets and generate/simulate additional data	Mahesh	Open
AI Model Accuracy	Medium	High	12	Model validation, tuning and diverse test images	Mahesh	Open
GPS Inaccuracy	Medium	Medium	9	Use simulated/test telemetry and tolerance zones	Jagesh Goyal	Open
API/Service Failure	Medium	Medium	9	Error handling, fallback logic and local testing	Jagesh Goyal	Open
Integration Issues	Medium	Medium	9	Modular development and integration testing	Both	Open

17. Research and Evaluation

17.1 Study of Existing Platforms

Existing valet and vehicle-management solutions mainly provide **vehicle check-in, digital tickets, tracking or inspection features**. However, a single platform combining **AI-based damage detection, security monitoring, suspicious movement detection and retrieval-time prediction** is less common. This gap motivated the development of our proposed system.

17.2 Evaluation Using Test Data

The system will be evaluated using **vehicle images, simulated GPS/telemetry data and historical/synthetic valet records**.

The evaluation will measure:

- Vehicle damage detection accuracy
- Geofence and suspicious movement detection
- Retrieval-time prediction error
- API/system response time

17.3 User Feedback

Potential users such as **vehicle owners and valet operators** can provide feedback on ease of use, dashboard clarity, trust in damage reports, security alerts and retrieval information. This feedback will be used to improve the system interface and functionality.

17.4 KPI Tracking

KPI	Target / Measurement
Damage Detection	≥ 85% accuracy
Security Detection	≥ 90% detection rate
Retrieval Prediction	≤ 5 minutes MAE
API Response	≤ 2 seconds
System Availability	≥ 99%
User Task Completion	≥ 80%

18. Appendices

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
CV	Computer Vision
GPS	Global Positioning System
IoT	Internet of Things
KPI	Key Performance Indicator
SLO	Service Level Objective
p95	95th percentile response time
Geofence	Virtual geographic boundary used for monitoring vehicle movement
RTO	Recovery Time Objective
RPO	Recovery Point Objective

18.1 References

For your project, use references related to **Computer Vision, YOLO/OpenCV, Machine Learning, GPS/Geofencing, and vehicle damage detection** rather than the stock-trading references in the original template.

Reference	Purpose
OpenCV Documentation	Computer vision and image processing
Ultralytics YOLO Documentation	Object detection and damage detection
Scikit-learn Documentation	Machine learning and evaluation
Python Documentation	Backend and data processing
Leaflet Documentation	Maps and geospatial visualization
FastAPI / Flask Documentation	Backend API development
PostgreSQL / MySQL Documentation	Database implementation
